



NORTH SYDNEY BOYS HIGH SCHOOL

2023 YEAR 11 ASSESSMENT TASK 3 YEARLY EXAMINATION 1st of September

Mathematics Extension 1

General Instructions

- Working time – 70 minutes + Reading time – 5 minutes
- Write in this booklet
- Write using blue or black pen
- NESA approved calculators may be used
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

Class Teacher: Please shade the circle.

- ☐ Mr Berry
- ☐ Ms Fu
- ☐ Mr Ireland
- ☐ Ms Lee
- ☐ Ms Moss
- ☐ Dr Vranešević

Student Number: _____

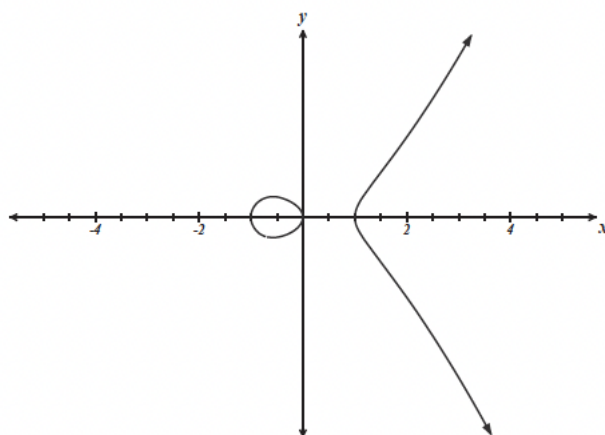
(To be used by the exam markers only.)

Section	A	B	C	D	E	Total	Total
Mark	$\overline{15}$	$\overline{14}$	$\overline{6}$	$\overline{20}$	$\overline{9}$	$\overline{64}$	$\overline{100}$

Section A (15 Marks)

1. The diagram shows the graph of $y^2 = f(x)$

1



Which expression best represents the function $f(x)$

- A) $x^2(1 - x)$
- B) $x^2(x - 1)$
- C) $x(1 - x^2)$
- D) $x(x^2 - 1)$
2. What is the natural domain of the function $f(x) = \frac{1}{2} (x \sqrt{x^2 - 1} - \log_e(x + \sqrt{x^2 - 1}))$?
- A) $x \geq 1$
- B) $x \leq -1$
- C) $-1 \leq x \leq 1$
- D) $x \leq -1$ or $x \geq 1$

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3. Solve for k in the following: $\frac{2k}{1-k} \geq k$ 3

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4. Find the Cartesian equation which is defined parametrically as 3

$$\begin{cases} x = \sqrt{t} \\ y = 2\sqrt{1-t} \end{cases}$$

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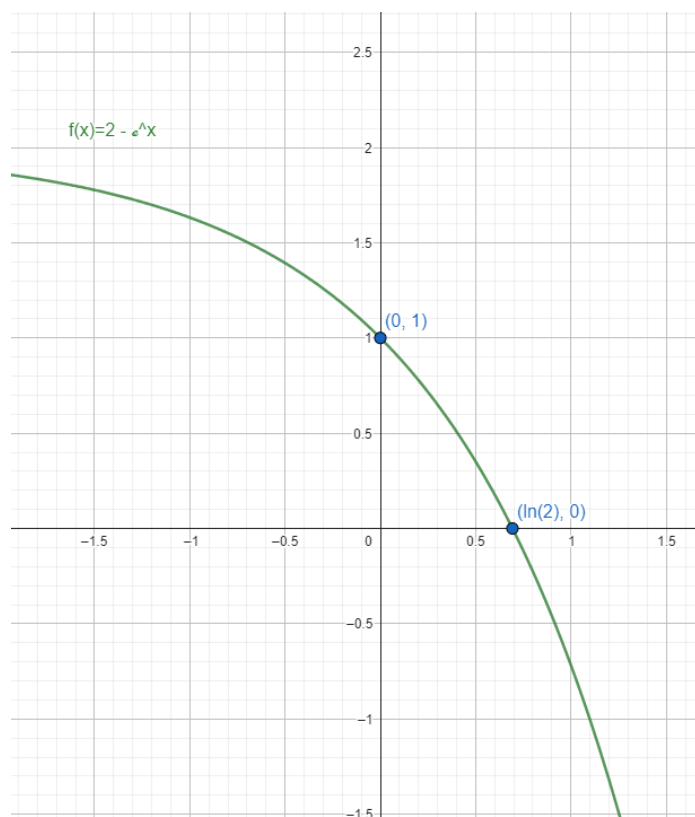
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5. The diagram shows the graph of a function $f(x) = 2 - e^x$.



Sketch the following functions **in the provided grid**, indicate all important features.

i) $y = (f(x))^2$

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ii) $y = \frac{1}{f(x)}$

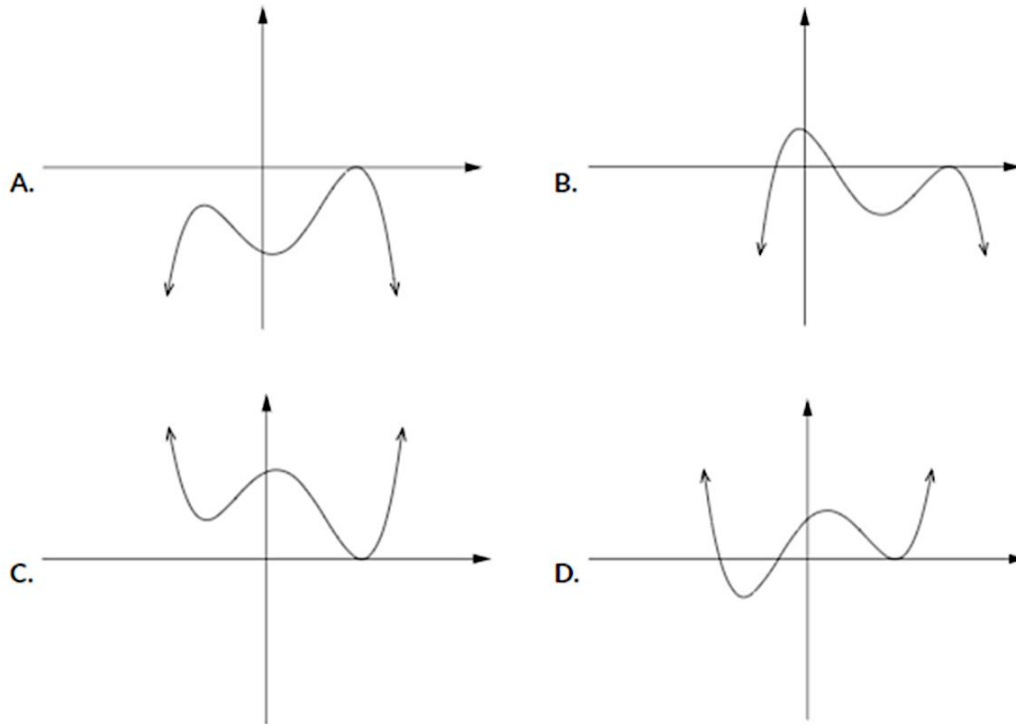
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iii) $y^2 = |f(x)|$

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Section B (14 Marks)

1. A monic polynomial $p(x)$ of degree 4 has one repeated zero of multiplicity 2 and is divisible by $x^2 + x + 1$. Which of the following could be the graph of $p(x)$? 1



2. Show that $(x - 1)(x - 2)$ is a factor of $P(x) = x^n(2^m - 1) + x^m(1 - 2^n) + (2^n - 2^m)$ where m and n are positive integers. 2

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3. α, β and γ are solutions of $2x^3 - 4x^2 - 3x - 1$,

i) find the value of $\alpha\beta + \beta\gamma + \alpha\gamma$ 1

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ii) find the value of $(\alpha - 1)(\beta - 1)(\gamma - 1)$ 2

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iii) Hence or otherwise, find the value of $(\beta + \gamma - \alpha)(\gamma + \alpha - \beta)(\alpha + \beta - \gamma)$ 2

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4. The polynomial $P(x) = x^3 + ax^2 + b$ has a zero at r and a double zero at 4.

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Find the values of a , b and r .

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5. When a polynomial is divided by x , $(x - 2)$ and $(x + 3)$, the remainders are 1, 3 and 13 respectively. If the same polynomial is divided by $x(x - 2)(x + 3)$, the remainder is $g(x)$, then find the value of $g(1)$. 3

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Section C (6 marks)

1. Four female and four male athletes are arranged in a row for the presentation of prizes. In how many ways can this be done if the males and females must alternate? 1

- A) $4! \times 4!$
B) $4! \times 5!$
C) $2 \times 4! \times 4!$
D) $2 \times 4! \times 5!$

2. Four digit numbers are formed from the digits 1, 2, 3 and 4. Each digit is used once only. The sum of all the numbers that can be formed is? 1

- A) 6666
B) 44440
C) 66660
D) 266640

3. How many distinct arrangements are there of the letters of the word ERATOSTHENES? 2

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4. A child has the following six plastic magnetic numbers: 6,2,6,7,6,2 How many different values can be created by using only five of the magnetic numbers? 2

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Section D (20 marks)

1. Find the exact value of $\tan[\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) - \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)]$, show working out. 2

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2. Show that if $\tan X = \frac{7m+6}{7-6m}$ and $\tan Y = \frac{4m+3}{4-3m}$, the value of $\tan (X - Y)$ is independent of m . 3

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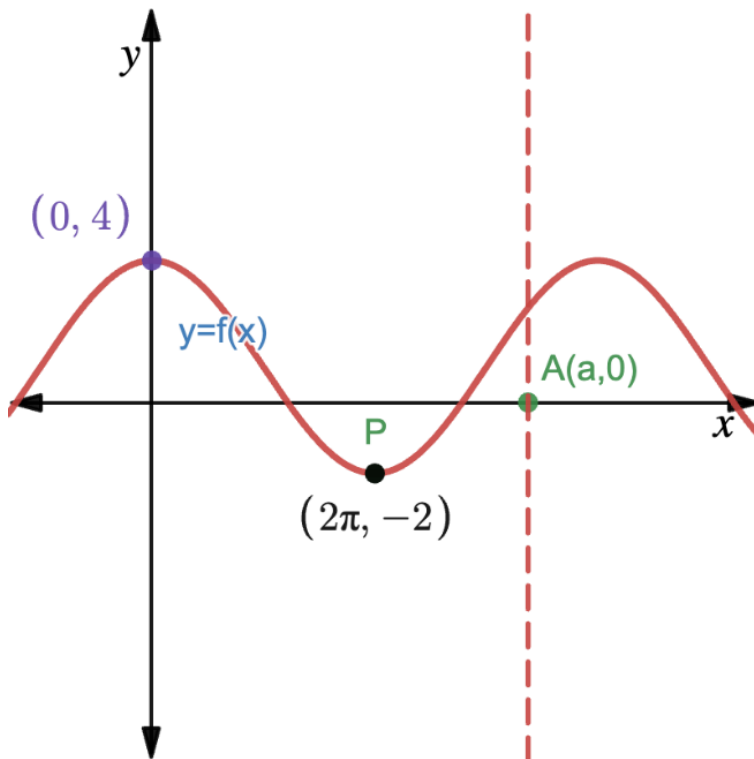
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3. Let $f(x) = 1 + 3 \cos \frac{x}{2}$. The diagram shows the graph of $y = f(x)$.



- i) What is the largest positive domain, containing $x = 0$, for which $f(x)$ has an inverse 1

function?

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- ii) Find the equation of $f^{-1}(x)$. 1

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iii) Sketch the curve $y = f^{-1}(x)$, showing all important features.

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iv) $A(a,0)$ lies to the right of $P(2\pi, -2)$ as indicated in the diagram. Find a simplified expression for the exact value of $f^{-1}(f(a))$.

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4. i) Use the substitution $t = \tan \frac{x}{2}$ to show that:

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$$2\sin x - 3\cos x = \frac{3t^2 + 4t - 3}{1 + t^2}$$

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- ii) Hence solve the equation $2\sin x - 3\cos x = 2$ for $0^\circ \leq x \leq 360^\circ$.

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Answer correct to the nearest degree.

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5. i) Use the fact that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$, solve $8x^3 - 6x - 1 = 0$ 2

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- ii) **Hence**, show that $\cos \frac{\pi}{9} = \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9}$ 2

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Section E (9 marks)

1.

Find the term in x^5 in the expansion of $\left(2x - \frac{1}{x^2}\right)^8$.

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2.

i) Show that $(1 + x)^m \left(1 - \frac{1}{x}\right)^m = \left(x - \frac{1}{x}\right)^m$

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- ii) By considering the term(s) independent of x in the expansion of the results from part i, 3

show that:

$$\binom{2002}{0}^2 - \binom{2002}{1}^2 + \binom{2002}{2}^2 - \dots + \binom{2002}{2002}^2 = -1 \binom{2002}{1001}$$

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iii) Hence, or otherwise, show that

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$$\sum_{k=0}^{1001} (-1)^k \binom{2002}{k}^2 = -\frac{1}{2} \binom{2002}{1001} \left[1 + \binom{2002}{1001} \right]$$

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End of Examination

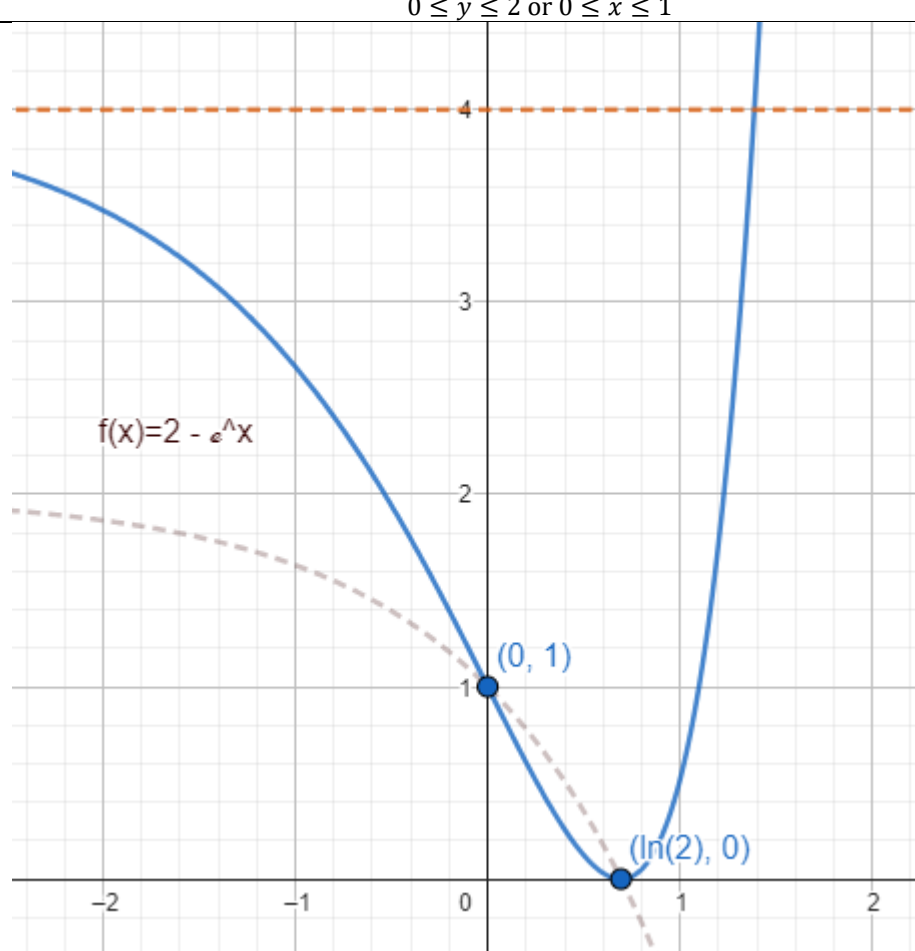
Extra Writing Space

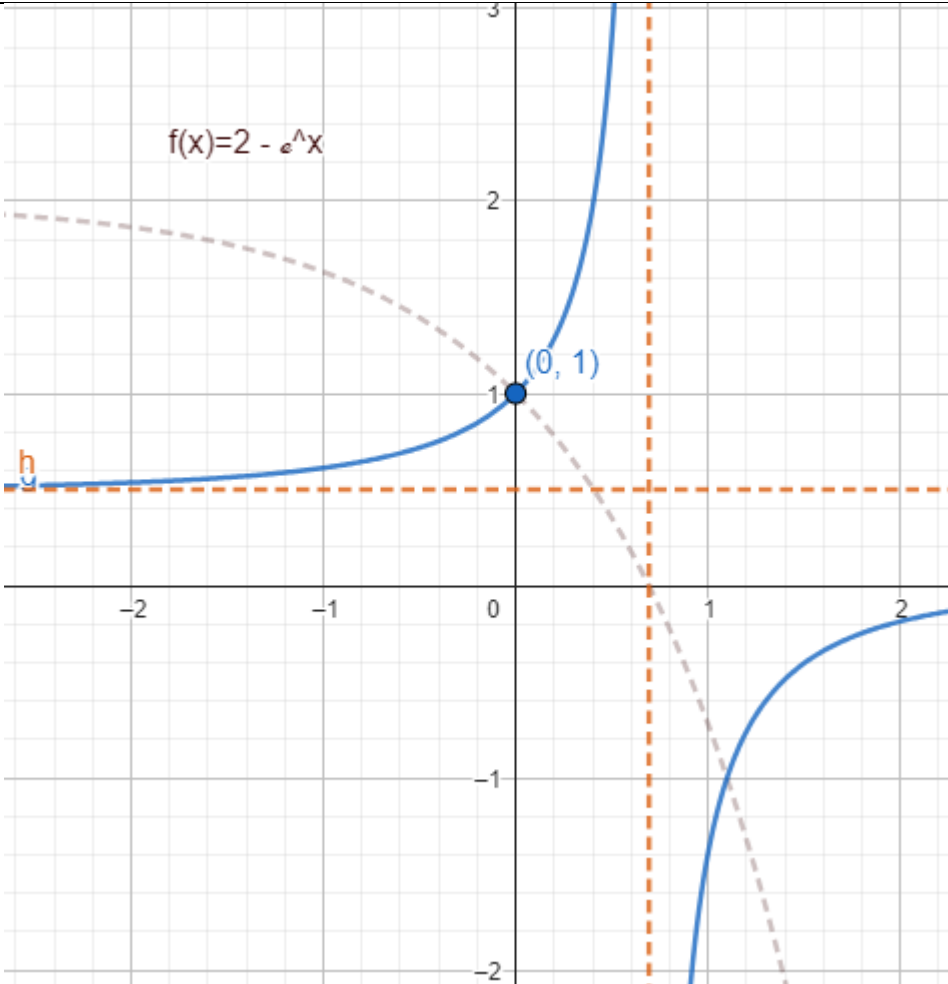
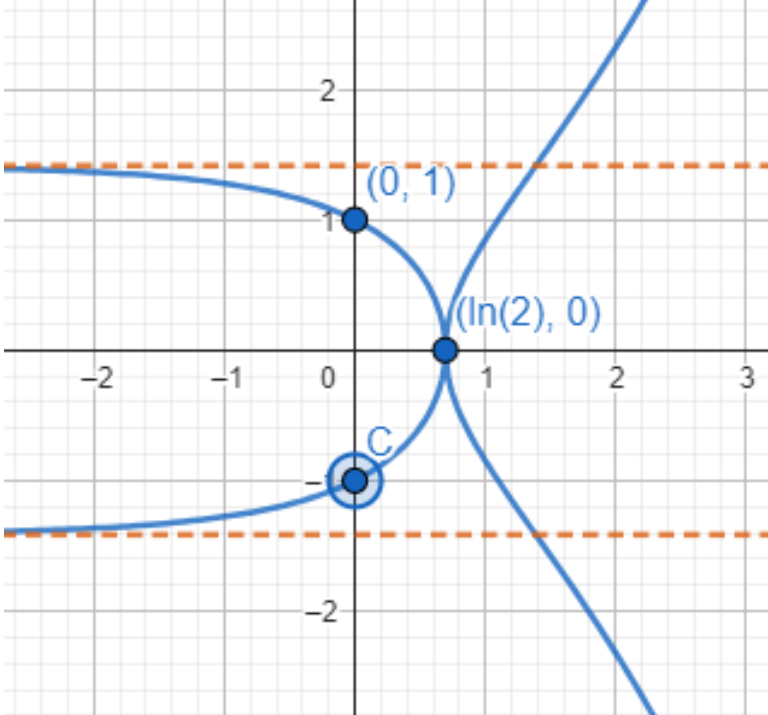
A large rectangular area for writing, containing 20 horizontal dotted lines for handwriting practice.

Extra Writing Space

A large rectangular area for writing, containing 20 horizontal dotted lines for handwriting practice.

Section A

	Suggested Solutions	Marks Awarded & comments
1.	D Zeros at $x = -1, 0$ and 1 , as $x \rightarrow \infty, y \rightarrow \infty$	1
2.	A $x^2 - 1 > 0$ and $x + \sqrt{x^2 - 1} > 0$, i.e. $x \leq -1, x \geq 1$; Thus, $x \geq 1$	1
3.	$k \neq 1$ $2k(1 - k) \geq k(1 - k)^2$ $2k(1 - k) - k(1 - k)^2 \geq 0$ $k(1 - k)(2 - 1 + k) \geq 0$ $k(1 - k)(1 + k) \geq 0$ $k \leq -1 \text{ or } 0 \leq k < 1$	1 1 1
4.	$x^2 = t$ $y^2 = 4(1 - t)$ $y^2 = 4 - 4x^2$ Since $t \geq 0$ and $1 \geq 1 - t \geq 0$, $0 \leq y \leq 2$ or $0 \leq x \leq 1$	1 1 1 for $x \geq 0$
5 i	 $f(x) = 2 - e^x$ $(0, 1)$ $(\ln(2), 0)$	1 - $x \rightarrow -\infty, y \rightarrow 4$ asymptote 1 - two intercepts

5ii	 $f(x) = 2 - e^x$ $(0, 1)$ h	1 – both Asymptotes Note: $x \rightarrow \infty, y \rightarrow 0$ $x \rightarrow -\infty, y \rightarrow \frac{1}{2}$ 1 – shape and intercept
5iii	 $(0, 1)$ $(\ln(2), 0)$ C	1– all 3 intercepts 1- Shape of both graphs 1- Horizontal asymptotes

	Suggested Solutions	Marks Awarded & comments
1.	C Only two real roots, which means one x -intercept, and $x \rightarrow \pm\infty, y \rightarrow \infty$	1
2.	$P(x) = x^n(2^m - 1) + x^m(1 - 2^n) + (2^n - 2^m)$ $P(2) = 2^n(2^m - 1) + 2^m(1 - 2^n) + (2^n - 2^m)$ $= 2^n 2^m - 2^n + 2^m - 2^m 2^n + 2^n - 2^m$ $= 0$ Similar for $P(1) = 0$, thus $(x - 1)(x - 2)$ is a factor of $P(x) =$ $x^n(2^m - 1) + x^m(1 - 2^n) + (2^n - 2^m)$	1 1
3i.	$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a} = -\frac{3}{2}$	1
3ii.	$(\alpha - 1)(\beta - 1)(\gamma - 1)$ $= \alpha\beta\gamma - (\alpha\beta + \beta\gamma + \alpha\gamma) + (\alpha + \beta + \gamma) - 1$ $= \frac{1}{2} - \left(-\frac{3}{2}\right) + 2 - 1$ $= 3$	1 1
3iii.	Since $\alpha + \beta + \gamma = 2$, then $\alpha + \beta - \gamma = 2 - 2\gamma$ $\alpha - \beta + \gamma = 2 - 2\beta$ $-\alpha + \beta + \gamma = 2 - 2\alpha$ $(\beta + \gamma - \alpha)(\gamma + \alpha - \beta)(\alpha + \beta - \gamma)$ $= (2 - 2\gamma)(2 - 2\beta)(2 - 2\alpha)$ $= -8(\gamma - 1)(\beta - 1)(\alpha - 1)$ $= -8 \times 3$ $= -24$	1 1

	$5a = 5$ $a = 1$ $b = -1$ Thus $R(x) = x^2 - x + 1$ $g(1) = R(1) = 1^2 - 1 + 1 = 1$	1
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Section C

	Suggested Solutions	Marks Awarded & comments
1.	C 4! Arrangements for females, 4! Arrangements for males; number of arrangements for two groups: $2 \times 4! \times 4!$	1
2.	C The digit one can have a value of $(1000 + 100 + 10 + 1) = 1111$, with the other numbers arranged 3! Ways. The value of all numbers is $3! \times 1111 \times (1 + 2 + 3 + 4) = 66660$	1
3.	ERATOSTHENES has 12 letters, with 3 E's, 2 T's, 2 S's $\frac{12!}{3!2!2!} = 19,958,400$	2
4.	A group of five numbers out of 6, must contain either three 6's or two 6's. When three 6's are used, there's two ways to have 2's, either one 2 and one 7 or two 2s, 66622 or 66627: $\frac{5!}{3!2!} + \frac{5!}{3!} = 30$ 66227: $\frac{5!}{2!2!} = 30$ Total different numbers = $30+30=60$	1 1

Section D

	Suggested Solutions	Marks Awarded & comments
1.	$\tan[\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) - \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)]$ $= \tan\left(\pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)\right)$ $= \tan\left(\pi - \frac{\pi}{4} + \frac{\pi}{4}\right)$ $= \tan(\pi)$ $= 0$	1 - correct use of identities or of exact value results 1

2.	$\tan(X - Y) = \frac{\tan X - \tan Y}{1 + \tan X \tan Y}$ $= \frac{\frac{7m+6}{7-6m} - \frac{4m+3}{4-3m}}{1 + \frac{7m+6}{7-6m} \times \frac{4m+3}{4-3m}}$ $= \frac{(7m+6)(4-3m) - (4m+3)(7-6m)}{(7-6m)(4-3m) + (7m+6)(4m+3)}$ $= \frac{3m^2 + 3}{46m^2 + 46}$ $= \frac{3}{46}$	
3i.	$0 \leq x \leq 2\pi$	1
3ii.	$x = 1 + 3 \cos \frac{y}{2}$ $\cos \frac{y}{2} = \frac{1}{3}(x - 1)$ $y = 2 \cos^{-1}\left(\frac{1}{3}(x - 1)\right)$	1
3iii.		1 - Two end points 1 – shape and 1 other point (y-intercept or $(1, \pi)$) OR 1- Correct domain & Range 1 – some correct points

	Thus, $\tan \frac{x}{2} = 1$ $\frac{x}{2} = \tan^{-1}(1)$ $\frac{x}{2} = \frac{\pi}{4}$ $x = \frac{\pi}{2} = 90^\circ$ or $\tan \frac{x}{2} = -5$ related angle $\frac{x}{2} = \tan^{-1} 5$ $\frac{x}{2} \approx 101.3099 \dots$ $x \approx 202.62$ Thus, $x = 90^\circ, 203^\circ$	2
5i	Let $x = \cos \theta$, then $8(\cos^3 \theta) - 6(\cos \theta) - 1 = 0$ $2(\cos 3\theta) = 1$ $\cos 3\theta = \frac{1}{2}$ $\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$ Only taken the first three solutions as it's a cubic function. Thus, $x = \cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$	1 1

5ii	As x is the roots of the cubic function, then $\cos \frac{\pi}{9} + \cos \frac{5\pi}{9} + \cos \frac{7\pi}{9} = 0$ $\cos \frac{\pi}{9} = -\cos \frac{5\pi}{9} - \cos \frac{7\pi}{9}$ $\cos \frac{\pi}{9} = \cos \left(\pi - \frac{5\pi}{9} \right) + \cos \left(\pi - \frac{7\pi}{9} \right)$ $\cos \frac{\pi}{9} = \cos \frac{4\pi}{9} + \cos \frac{2\pi}{9}$ $\cos \frac{\pi}{9} = \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} \text{ as required}$	1 1
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Section E

	Suggested Solutions	Marks Awarded & comments
1.	General term of $\left(2x - \frac{1}{x^2}\right)^8$ is $\binom{8}{i}(2)^{8-i}(x)^{8-i}(-1)^i(x)^{-2i}$, $T_k = \binom{8}{i}(-1)^i(2)^{8-i}(x)^{8-3i}$ For x^5, $5 = 8 - 3i$ $i = 1$ Thus, the term in x^5 is $\binom{8}{1}(-1)^1(2)^{8-1}x^5 = -1024x^5$	1 1
2i.	LHS = $(1+x)^m \left(1 - \frac{1}{x}\right)^m$ $= \left((1+x) \left(1 - \frac{1}{x}\right) \right)^m$ $= \left(1 - \frac{1}{x} + x - 1 \right)^m$ $= \left(x - \frac{1}{x} \right)^m$ = RHS	1

